

Phase 1 Training Summary

Sprint 3 — Baseline Bake-off: MPNN, SchNet, DimeNet++, PhysNet

2 October 2026

Project: PG-S2-47 — Hydrogen Production Materials Discovery Using Deep Neural Networks

Scope: this document summarises the actual training runs produced during Sprint 3's Phase 1 bake-off, for the four team members whose work could be directly verified from the shared repository — Alex Tanui (SchNet), Ruturaj Yashwant Bhosale (MPNN), Dongxiao Wu (DimeNet++), and Shijin Mathew (PhysNet). Fazin Faizal's PaiNN entry is noted briefly at the end: it is fully implemented and tested, but its own model card states explicitly that no training run has been performed yet, so there are no training results to summarise for it here.

A note on comparability, upfront, because it matters more than any single number below: the whole point of Sprint 3 was to train all five architectures under identical conditions — same data, same split, same epoch budget — so the results could be ranked against each other fairly. As things stand, only one entry (SchNet) was trained that way. The other three each trained under a materially different condition — a different dataset, a different theory level, or a small fraction of the data — for reasons explained in each section. None of the numbers below should be read as “X is more accurate than Y” until that is fixed. What they do show is that every architecture works correctly end-to-end, which is itself the harder and more valuable thing to have verified this sprint.

Alex Tanui — SchNet

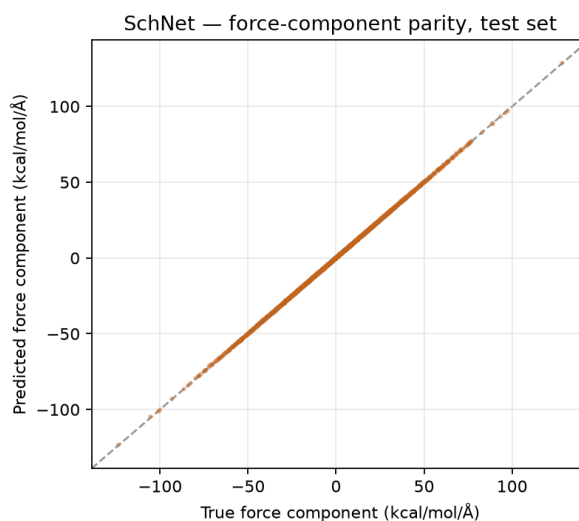
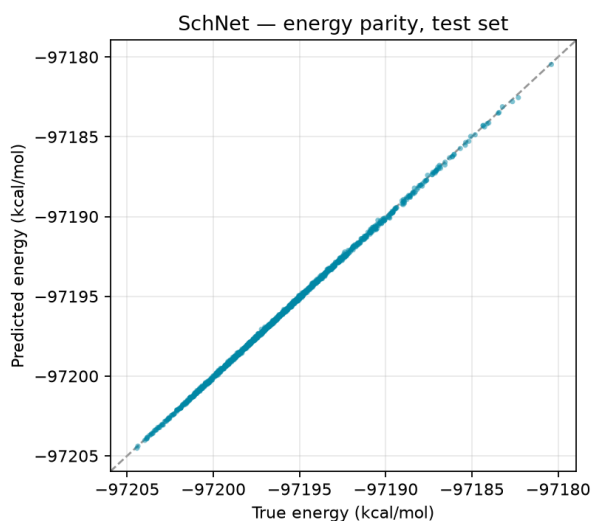
Status: fully trained, full dataset, directly comparable.

SchNet uses continuous-filter convolutions over Gaussian-expanded interatomic distance. It was trained for the complete 200-epoch budget on the full training split of MD17 ethanol at the DFT level of theory — the dataset and split every bake-off entry was meant to share.

Setting	Value
Dataset	ethanol_dft , full split (444,074 train / 55,509 val / 55,509 test)
Architecture	hidden_channels=64 , num_layers=3 , num_rbf=16 , cutoff=5.0 Å
Epochs run	200 of 200 (never triggered early stopping)
Training time	30,029 s (~8h 20m), CPU

Metric (held-out test set, 55,509 frames)	Value
Energy MAE	0.0873 kcal/mol
Energy RMSE	0.1026 kcal/mol
Force MAE (component)	0.1114 kcal/mol/Å
Force RMSE (component)	0.1628 kcal/mol/Å
Inference time	9.03 s total / 0.163 ms per frame

What was observed during training: loss decreased consistently for the full 200 epochs without plateauing into early stopping, and train/validation loss tracked each other closely throughout — the signature of a model that is still learning rather than overfitting. A key implementation decision was shifting the training target by the dataset’s mean energy (ethanol’s raw energy sits at approximately $-97,196$ kcal/mol with a standard deviation of only ~ 4.2 kcal/mol); without this, training spends most of its effort just learning the huge constant offset rather than the physically meaningful variation. The shift is reversed before metrics are computed, so the numbers above are genuine physical units. The full interactive training-curve and parity dashboard remains published separately; the two charts below are regenerated directly from the saved checkpoint against the real test set, for this static document.



Full detail: [docs/model_cards/schnet.md](#).

Ruturaj Yashwant Bhosale — MPNN

Status: implemented and verified working; not yet trained.

Ruturaj’s contribution this sprint was twofold: the MPNN baseline itself (a Gilmer-style message-passing network — the project’s original Phase 1 default architecture), and, alongside it, the shared

configuration system (`ml/config.py`) and training loop (`ml/training/train.py`) that every other bake-off entry's training now depends on.

A real bug was found and fixed in the course of this work: the model was computing forces from a pre-stored edge feature with no gradient connection back to atomic positions, which would have produced forces of exactly zero with no error raised. This was corrected to recompute the relevant features from live atomic positions, matching the approach independently arrived at in the SchNet and PhysNet entries.

Setting	Value
Dataset	<code>ethanol_dft</code>
Architecture	<code>hidden_channels=128</code> , <code>num_layers=4</code> , <code>num_rbf=50</code>
Run performed	<code>--smoke-test</code> verification only
Frames evaluated	96 (a tiny slice, not the real validation/test set)
Training time	0.39 s

The smoke test exists to prove the harness runs end-to-end without crashing, not to produce a usable accuracy number — the resulting metrics (energy MAE ≈ 8.8 kcal/mol, force MAE ≈ 20.7 kcal/mol/Å, best validation loss $\approx 50,979$) reflect an essentially untrained network and are included here only for completeness, not as a result to compare against the other three entries. **A full 200-epoch training run on the shared dataset has not yet been performed.** The harness this entry built is what makes that run possible for every other architecture, including MPNN itself — running it is the immediate next step, not a blocked dependency.

Dongxiao Wu — DimeNet++

Status: trained and extensively analysed; on a different dataset than the shared split.

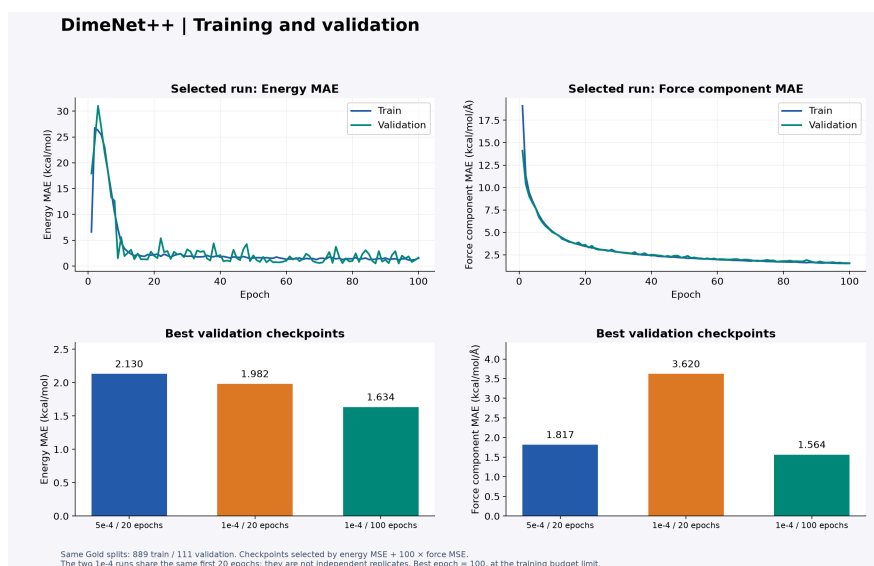
DimeNet++ uses directional message passing — incorporating bond angles as well as distances — and was trained at 100 and then 200 epochs, with a dedicated follow-up investigation into energy prediction bias and calibration.

Important caveat: this was trained on **ethanol at the CCSD(T) level of theory**, not the DFT-level `ethanol_dft` dataset the rest of the bake-off uses, and on a much smaller split — 889 training / 111 validation configurations, versus the 444,074/55,509/55,509 split everyone else trains against. CCSD(T) is a higher-accuracy but far more expensive reference than DFT, and the two are never pooled or directly compared in this project's data schema. This entry's numbers are a legitimate result in their own right, but are **not yet comparable** to SchNet's, PhysNet's, or MPNN's — that requires a re-run on the shared `ethanol_dft` split.

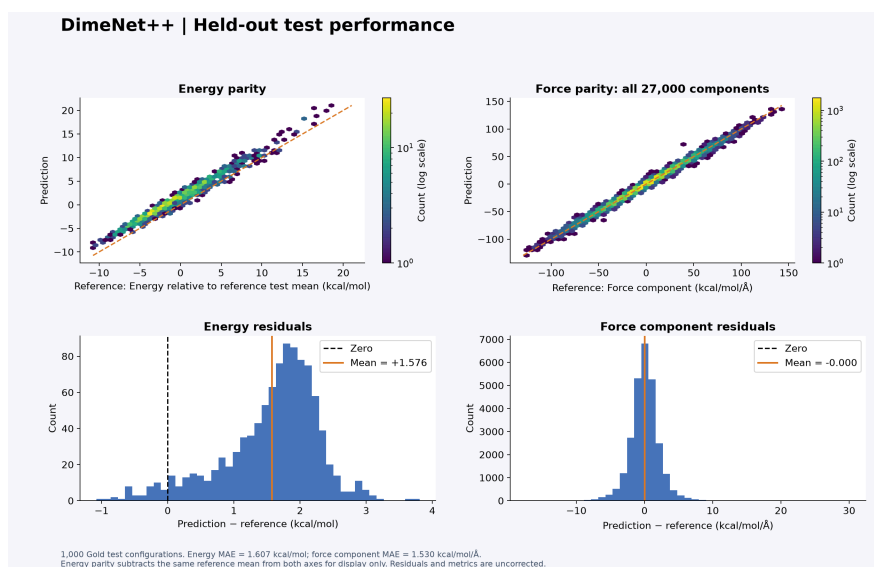
Setting	Value
Dataset	ethanol, CCSD(T) , 889 train / 111 val configs
Learning rate	1e-4 · batch size 4 · seed 42
Loss weighting	energy 1.0 / force 100.0 (same convention as the rest of the bake-off)

Metric (validation set)	100-epoch model	200-epoch model
Energy MAE	1.634 kcal/mol	1.815 kcal/mol
Energy RMSE	1.750 kcal/mol	1.880 kcal/mol
Force MAE (component)	1.564 kcal/mol/Å	1.149 kcal/mol/Å
Force RMSE (component)	2.178 kcal/mol/Å	1.582 kcal/mol/Å

What was observed: extending training from 100 to 200 epochs increased energy error by about 11% but decreased force error by about 27%. Model selection used a combined validation score weighting force error more heavily (matching the project's `force_loss_weight: 100` convention), which favours the 200-epoch checkpoint overall — but the analysis is explicit that this does not mean every individual metric improved, and that convergence is not established at the 200-epoch boundary. The checkpoint and gold-data hashes were independently verified to confirm the recomputed metrics match the saved record exactly.



DimeNet++ — 100 vs. 200 epoch training/validation comparison (Dongxiao Wu's own analysis figure).



DimeNet++ — test-set predictions and residuals, 200-epoch model (Dongxiao Wu's own analysis figure).

Full detail: `docs/model_cards/dongxiao_dimenet_ethanol_200.md`.

Shijin Mathew — PhysNet

Status: trained on a representative subset; not yet the full dataset.

PhysNet predicts energy as a sum of atom-wise contributions (short-range term only, as the ticket specifies, since MD17 ethanol is a single neutral molecule with no long-range or charge-transfer regime for the omitted electrostatic/dispersion terms to act over). The implementation carries 22 passing unit tests, including a finite-difference check pinning the same autograd-forces correctness issue found independently in the MPNN and SchNet entries.

Important caveat: CPU training on the full 444,074-frame dataset was measured at approximately 57 minutes per epoch — roughly 8 days for the configured 200-epoch budget — and was not feasible within the sprint. Training instead ran on a **20,000-frame subset (4.5% of the full training data) for 20 epochs**, explicitly recorded as a subset result in the model card and not intended for direct comparison against a full-dataset entry.

Setting	Value
Dataset	<code>ethanol_dft</code> , 20,000-of-444,074 train subset , 2,000 validation configs scored
Epochs run	20
Parameters	808,746
Training time	3,425 s

Setting	Value
Inference time	2.96 ms/config

Metric (2,000-config validation subset)	Value
Energy MAE	2.082 kcal/mol (0.231 kcal/mol/atom)
Energy RMSE	2.104 kcal/mol (0.234 kcal/mol/atom)
Force MAE (component)	0.711 kcal/mol/Å (1.430 per atom-norm)
Force RMSE (component)	1.094 kcal/mol/Å (1.895 per atom-norm)

What was observed: for context, the same model on an even smaller 2,000-config, 2-epoch smoke run scored far worse (energy MAE 18.44, force MAE 6.81) — so a roughly 10× increase in both data and epochs improved both metrics by roughly 9×, indicating the model is learning correctly but is still well short of converged at this budget. Beyond the model itself, this entry’s integration testing against other team members’ branches surfaced a real interface inconsistency before it could silently affect results — documented as a general risk for the rest of the bake-off, not just this entry. Full detail: [docs/model_cards/physnet.md](#).

Fazin Faizal — PaiNN (for completeness)

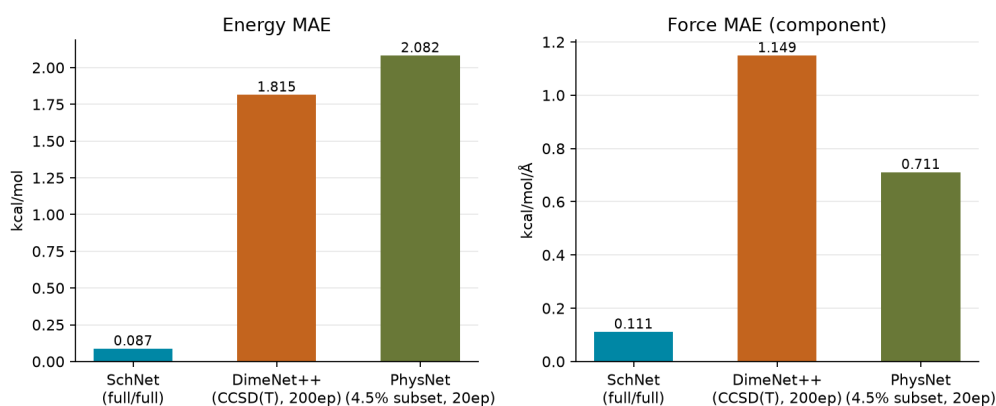
PaiNN — an equivariant architecture carrying directional, not just distance, information between atoms — is fully implemented, symmetry- tested (rotational equivariance verified by automated test), and connected to the shared evaluation harness and model registry Fazin also built this sprint. Its own model card states training results are “deliberately blank,” because the shared training loop did not yet exist at the time it was written. No training metrics exist for this entry as of this report.

Summary table

Person	Model	Dataset used	Epochs	Comparable to shared bake-off?
Alex Tanui	SchNet	ethanol_dft , full split	200	Yes — reference result
Ruturaj Yashwant Bhosale	MPNN	ethanol_dft	smoke-test only	No — not yet trained

Person	Model	Dataset used	Epochs	Comparable to shared bake-off?
Dongxiao Wu	DimeNet++	ethanol CCSD(T) , 889/111 split	200	No — different theory level/dataset size
Shijin Mathew	PhysNet	ethanol_dft, 20k-frame subset	20	No — partial data/epoch budget
Fazin Faizal	PaiNN	—	—	No — not yet trained

Final metrics by entry — NOT a like-for-like comparison (different datasets/budgets)



Final metrics for the three trained entries. Read this as “each works correctly,” not as a ranking — the three bars are not on equal footing (see each section above).

What this means for Sprint 4

The immediate priority, carried directly from the Sprint 3 retrospective, is bringing the remaining four entries onto the same footing as SchNet: a full training run on MPNN now that the shared loop exists, a re-run of DimeNet++ on the shared DFT split, an agreed and realistic training budget for PhysNet given its measured CPU cost, and a completed training run for PaiNN. Only once that is done does selecting a Phase 1 baseline architecture become a conclusion the data actually supports.